

old model average accuracy = 90%
 → 100 dataset
 Linear accuracy = 92%
 population SD = 5%

Hypothesis $\left. \begin{array}{l} H_0: \mu = 90 \\ H_1: \mu > 90 \end{array} \right\}$

calculate $z = \frac{92-90}{5/\sqrt{100}} = \frac{2}{0.5} = 4$

1.645 critical cut-off z score

Reject H_0 $z(4) > 1.645$

$\alpha = 0.05$

$1 - 0.05 = 0.95$

$P(Z < 1.645) = 0.95$
 $P(Z > 1.645) = 0.05$

Case sample size is small and std dev is unknown.

T-test

old model = 90%

→ 100 dataset

$\{92, 91, 93, 89, 94, 92, 90, 91, 95, 93\} \rightarrow \bar{x} = 92$
 $n = 10$ $s = 1.8$

Null Hypo $H_0: \mu = 90$

alternative hyp $H_1: \mu > 90$

$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

→ population SD
 $s \rightarrow$ sample SD

$t = \frac{92-90}{1.8/\sqrt{10}} = \frac{2}{0.56} = 3.4$

$t = 3.4$

$df = n-1 = 9$

$\alpha = 0.05$
 $df = 9$

$t_{\text{table}} = 1.83$

$z\text{-test} = 1.645$

$t(3.4) > 1.83$

Reject H_0

Statistical Tests:

fraud detection model:

→ fraud/not fraud

	Actual	Predicted
A	1	1
B	1	1
C	1	1
D	1	1
E	1	1

Model A → 91%
 Model B → 93%

Accuracy ≠ correctness

≠ evidence

Problems

- Compare old vs new recommendation engine
- Check if features affect target
- Compare model performances
- Compare 3+ models
- Validate assumptions
- Feature selection
- Model A's effectiveness

Test

- A/B Testing
- Chi Square test
- T-test
- ANOVA
- Z-tests
- chi square/anova
- Hypothesis testing

$H_0: \mu_1 = \mu_2$
 $H_1: \mu_1 \neq \mu_2$

$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

$t_{\text{table}} = 1.83$
 $t_{\text{calc}} = 4.16$
 $4.16 > 1.83$
 Reject H_0

Z-test

5000 → spend
 → average

→ 100 customers → sample average = 8300

$n = 100$ $\bar{x} = 8300$

→ population

GI

$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
 $\bar{x} \rightarrow$ sample mean
 $\mu \rightarrow$ population mean
 $\sigma \rightarrow$ population std dev.
 $n \rightarrow$ sample size

population mean = 100 (μ)
 popn. SD = 15 (σ)
 sample size = 36 (n)
 $\bar{x}$ mean = 106 ($\bar{x}$)
 $\rightarrow \frac{6}{2.5} = 2.4 \rightarrow z \text{ score}$

The sample mean is 2.4 std dev above the population mean.